Exp No: 10a
Date: 12/9/26 Internetworking with Routers

AIM:

To implement internetworking with routers on CISCO packet Tracer simulator.

Designing and configuring a simple internetwork:

In this network, a router and 2 PC's are used. Computers are connected with router using a copper straight-through cable.

Procedure:-

Step 1:- configuring Router

1. Select the router and open CLI
2. Press enter to start configuring the network
3. Type enable to activate the privilised mode.

Command for Router:-

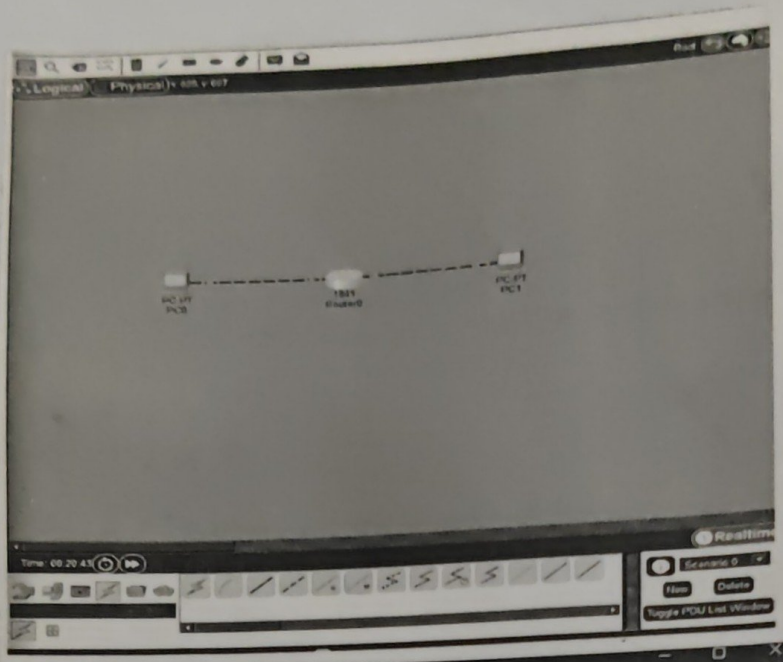
enable
config it
interface Fast Ethernet 0/10
ip address 192.168.191.255.255.255.0
no shutdown
interface Fast Ethernet 0/1
ip address 192.168.20.1.255.255.255.0
no shutdown

Step 2:- configuring PC's

1. Assign IP address to every PC
2. Select the PC, go to the desktop and select IP configuration and assign an IP address Default Gateway, Subnet mask.
3. Assign the default gateway of PC0 and PC1 as 192.168.10.1 and 192.168.20.1 respectively

Step 3:- Connecting PC's with router.

1. Connect the Fast Ethernet 0 port of PC with Fast Ethernet 0/0 port of Router 1
2. connect the Fast Ethernet 0 port of PC 1 with Fast Ethernet 0/1 port of Router



Router0

Physical Config CLI Attributes

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

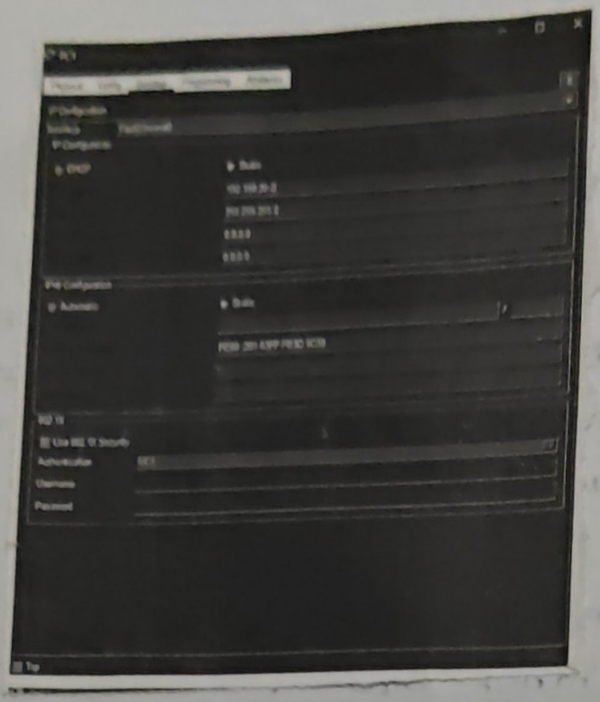
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    down
FastEthernet0/1        192.168.20.1    YES manual up                    down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

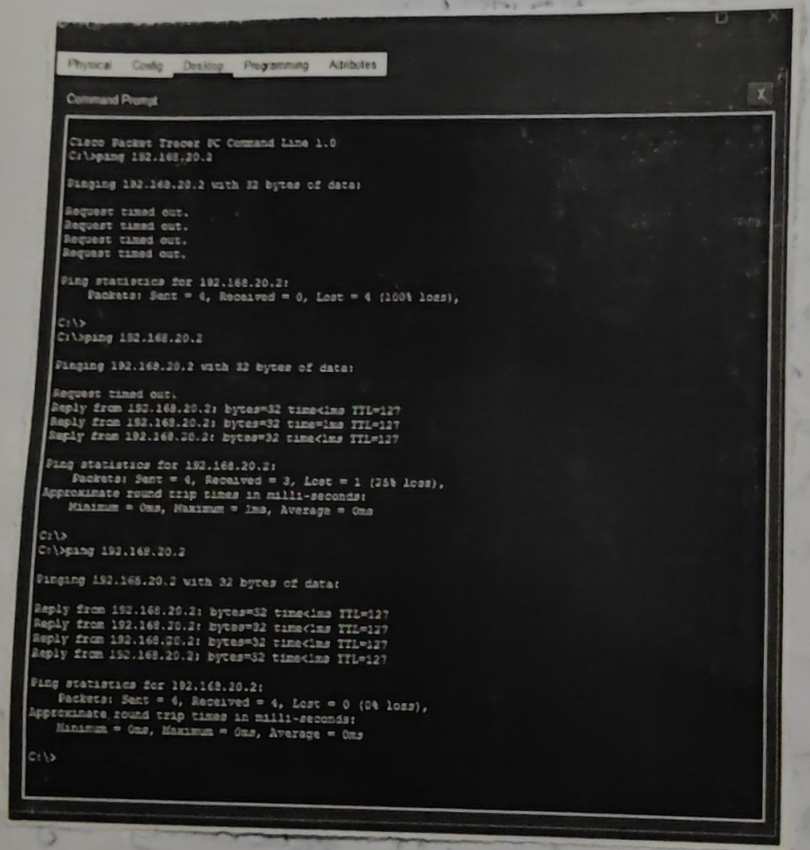
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Copy Paste

Top




Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

Result:

Thus internetworking with routers in CISCO Packet Tracer simulator was implemented Successfully.

Exp no: 10 b
Date: 12/9/26 Design and Configure an internetwork using wireless router, DHCP server and Internet cloud.

AIM :-

To design and configure an internetwork using wireless router, DHCP server and internet cloud.

Addressing Table :-

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	Ethernet 0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	Internet	DHCP		
Cisco.com Server	Ethernet 0	208.67.220.220	255.255.255.0	
Laptop	Wireless 0	DHCP		

Procedure :-

Part 1 :- Building a Simple network :-

Step 1 :- Build the topology

1, Add network devices using the device selection box into the workspace

2, Change the display name of the devices by clicking on the device icon and click on the config tab and type in the new name.

3, Add the physical cabling between devices.

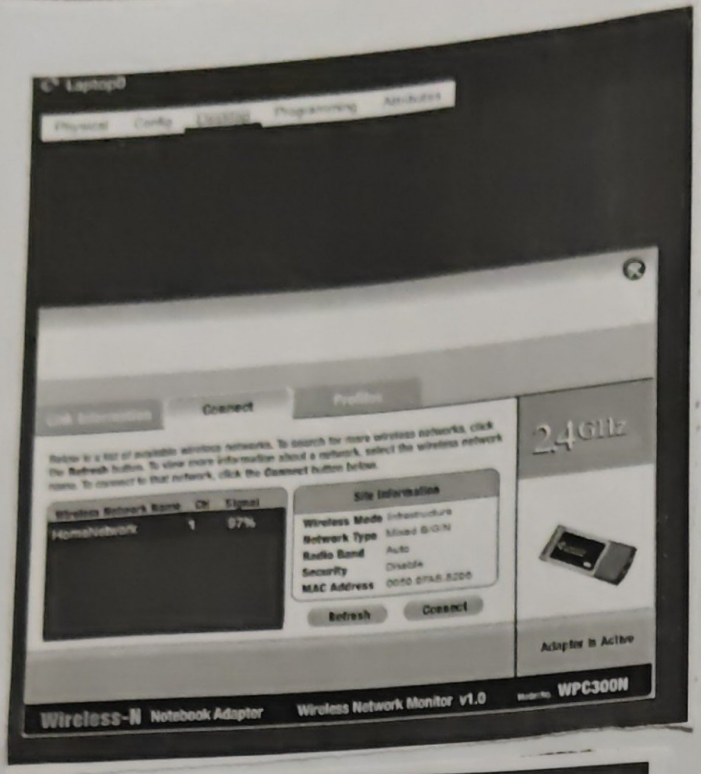
Part 2 :- Configure the Network Devices :-

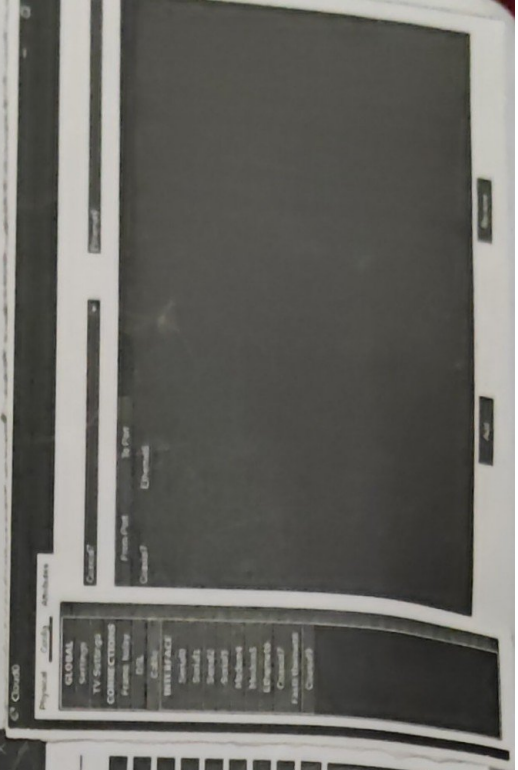
Step 1 :- Configure the wireless router

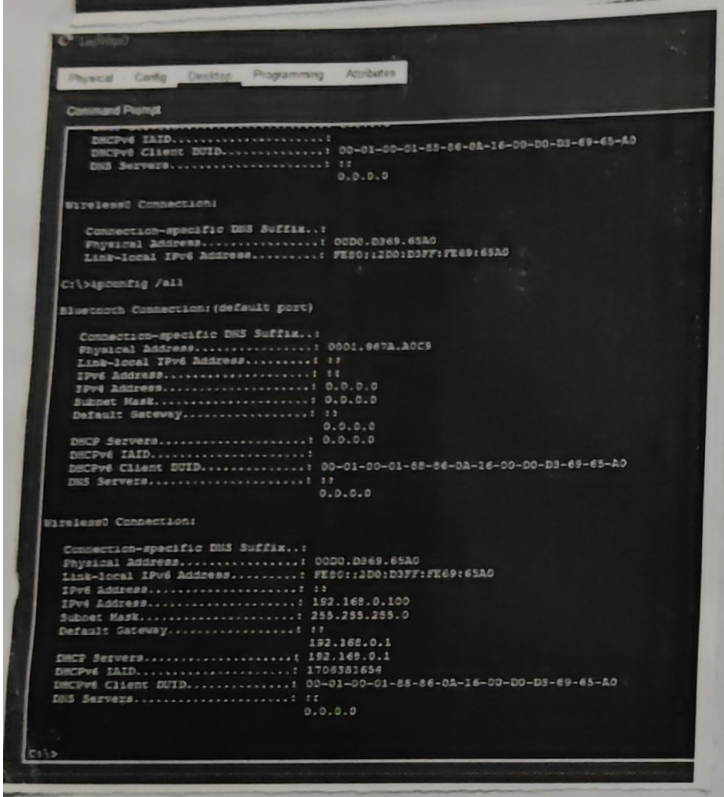
1, Click on the wireless router icon and click on GUI tab to view configuration options.

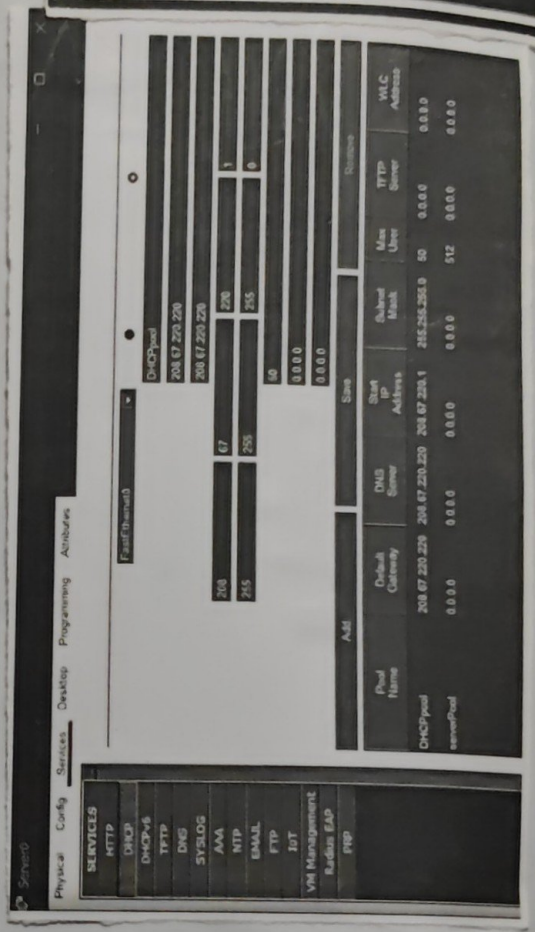
2. Then click on the wireless tab and the only thing that needs to be changed is network name (SSID).

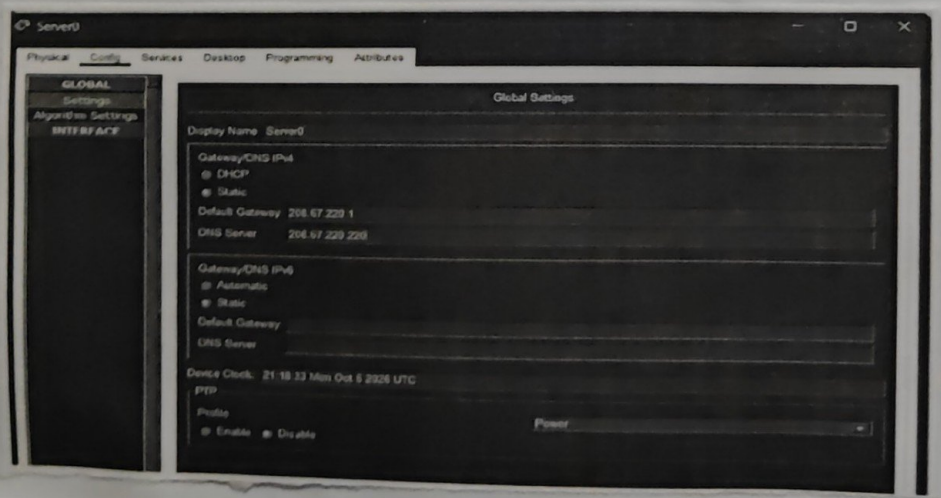
3, Then save the settings

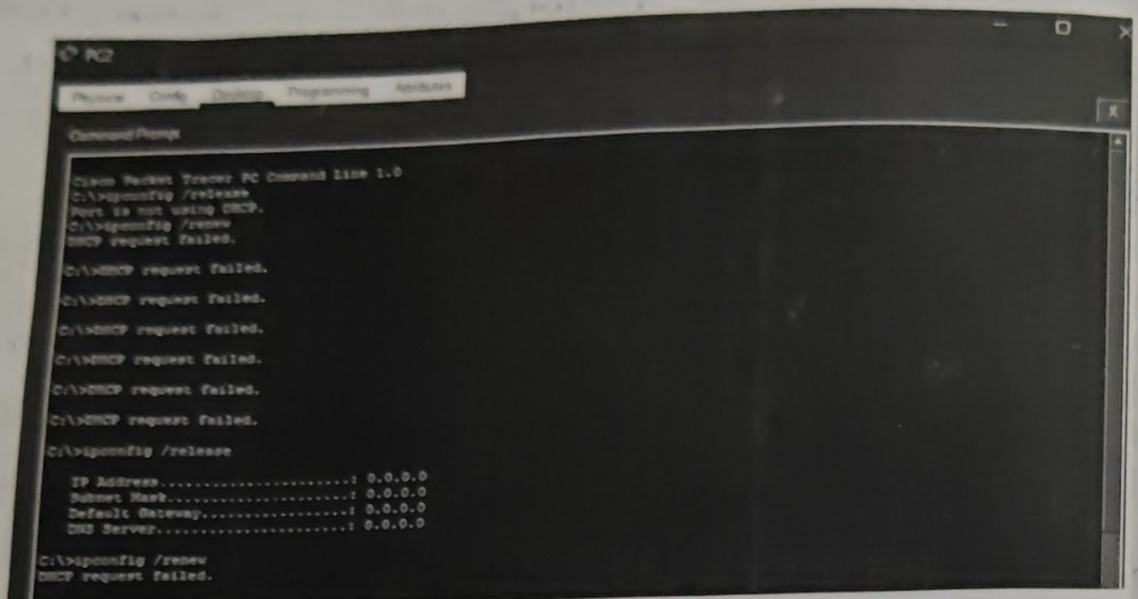

Laptop0
Physical
Config
Desktop
Programming
Attributes
Connect
Profiles
2.4GHz
Wireless Network Name
CH
Signal
HomeNetwork
1
97%
Site Information
Wireless Mode
Network Type
Radio Band
Security
MAC Address
Refresh
Connect
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor V1.0
WPC300N

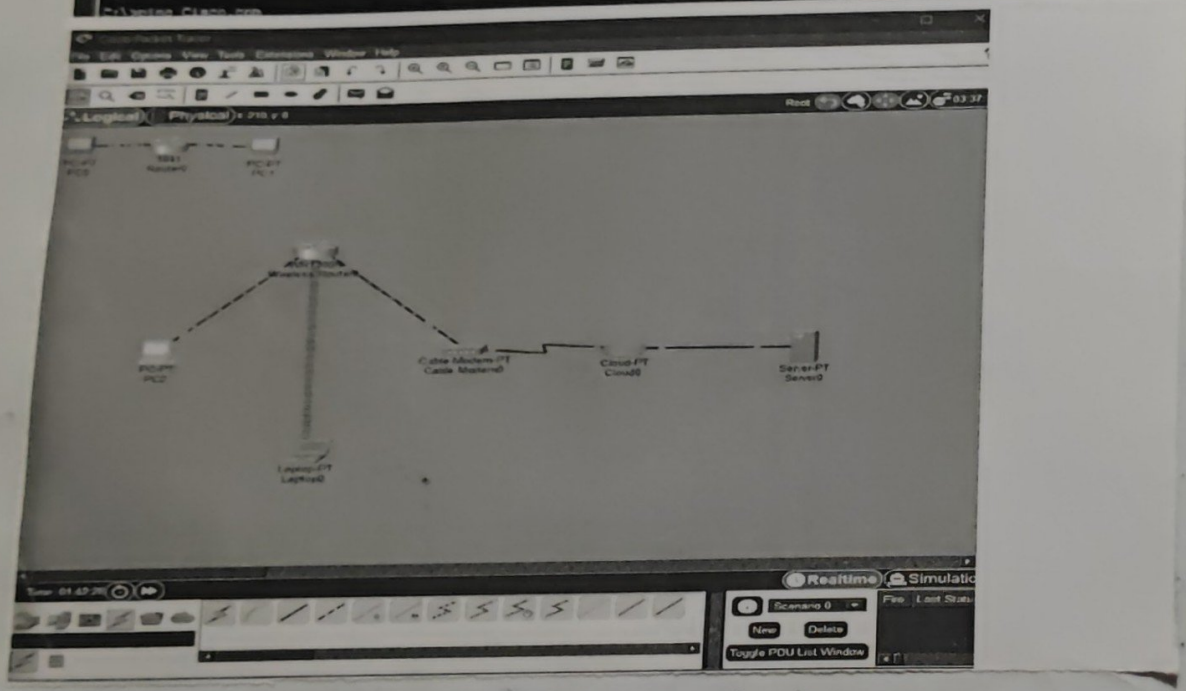



Physical
Config
Desktop
Programming
Attributes
Command Prompt
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-85-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Physical Address.....: 00D0.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
C:\>ipconfig /all
Bluetooth Connection:(default port)
Connection-specific DNS Suffix..:
Physical Address.....: 0001.967A.A0C9
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-85-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Physical Address.....: 00D0.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.0.1
DHCP Servers.....: 192.168.0.1
DHCPv6 IAID.....: 1706381654
DHCPv6 Client DUID.....: 00-01-00-01-85-86-0A-16-00-D0-D3-69-65-A0
DNS Servers.....: ::
0.0.0.0
C:\>


Physical
Config
Services
Desktop
Programming
Attributes
FastEthernet0
DHCPpool
208.67.220.220
208.67.220.220
208
255
67
255
236
255
1
0
40
0.0.0.0
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
DHCPpool
208.67.220.220
208.67.220.220
208.67.220.1
255.255.255.0
50
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
0.0.0.0
0.0.0.0
512
0.0.0.0
0.0.0.0
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PPP


Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Global Settings
Display Name Server0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 208.67.220.1
DNS Server 208.67.220.220
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock: 21:18:33 Mon Oct 5 2026 UTC
PPP
Profile
Enable
Disable
Power


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew
DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>ipconfig /release
IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0
DNS Server.....: 0.0.0.0
C:\>ipconfig /renew
DHCP request failed.


Logical
Physical
Root
PC-PT
PC2
Cable Modem-PT
Cable Modem0
Cloud-PT
Cloud0
Server-PT
Server0
Laptop-PT
Laptop0
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window

Step 2 :- Configure the Laptop:

1. Configure the laptop by clicking on its icon and then selecting physical tab.
2. There remove the Ethernet copper module and replace it with the wireless WPC 300N module

Step 3 :- Configure the PC.

1. Configure the PC for wired network

Step 4 :- Configure the Internet cloud.

1, click on the Internet cloud icon and then click on the Physical tab.
2. The cloud device needs 2 modules, PT-CLOUD-NM-1CX and PT-CLOUD-NM-1CFE
3. Identify the form and to ports and add them
4. In the config tab click Ethernet under Interface, select cable as the provider.

Step 5 :- Configure the cisco.com server

1, Configure the cisco.com server as a DHCP server.
2, Configure the server as DNS server to provide domain name to IPV4 address resolution.
3, Configure the Global settings in the config table.
4. Configure the Fast Ethernet 0 interface settings in the server.

Part 3: Verify the connectivity :-

1, verify that the PC is receiving IPV4 configuration from DHCP.
2. Test connectivity to the server from the PC

Result :-

Thus, the wireless router, DHCP server and Internet cloud were designed and configured with internetwork successfully.